THE UNITED REPUBLIC OF TANZANIA
ORGANIZATION OF HEADS OF NON-GOVERNMENT SECONDARY SCHOOLS-TANZANIA (OHONGSS-T)
MWANZA & KIGOMA REGIONS
FORM FOUR MOCK EXAMINATION-MAY, 2023

032/1 **CHEMISTRY 1**
Time: 3 Hours **Thursday, 25th May, 2023 P.M**

Instructions

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1. This paper consists of sections **A, B** and **C** with a total of **eleven (11)** questions.
2. Answer **all** questions in section **A** and **B** and **two (2)** question from section **C.**
3. Section **A** and **C** carry **fifteen (16)** marks, section **B** carries **fifty-four (54)** marks and section **C** carries **thirty (30)** marks.
4. Non-programmable calculators may be used.
5. Cellular phones and any unauthorized materials are not allowed in the examination room.
6. Write your **Examination Number** on every page of your answer booklet(s)
7. The following constants may be used.

   Atomic masses: H = 1, C = 12, O = 16, Mg =24.

   Avogadro's number = $6.02x10^{23}$

   GMV at s.t.p = $22.4dm^3$

   1 Faraday = 96, 500 coulombs.

   1 litre = $1dm^3 = 1000cm^3$.

This paper consists of **7** printed pages

SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i)-(x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.

   (i) Which of these statements is true about isotopes of an element?
   - A. The number of protons is the same, but the number of neutrons is different.
   - B. The number of protons and neutrons is the same, but the number of electrons is different.
   - C. The number of neutrons is the same, but the number of protons is different.
   - D. The number of protons is the same, but electrons are added to the nucleus.
   - E. The number of neutrons and electrons are the same.

   (ii) Which of the following transforms a mechanical energy to an electric energy?
   - A. Hydroelectric power plant
   - B. Solar panel
   - C. An electric cooker
   - D. An electric motor
   - E. A motorcycle wheel

   (iii) Which oxides among the following are gaseous at room temperature?
   - A. Copper oxide and iron oxide
   - B. Carbon dioxide and copper oxide
   - C. Carbon dioxide and Sulphur dioxide
   - D. Sulphur dioxide and copper oxide
   - E. Iron oxide and carbon dioxide

   (iv) Steam and methane can react and produce hydrogen according to the following equation.

   $CH_{4(g)} + H_2O_{(g)} \longrightarrow CO_{(g)} + 3H_{2(g)}$

   The volume of methane needed to produce $150cm^3$ of hydrogen at the same conditions will be;
   - A. $50cm^3$
   - B. $25cm^3$
   - C. $100cm^3$

D. $75cm^3$
E. $150cm^3$

(v) Which of the following statements is not true about hydrogen gas?
A. Is a reducing agent.
B. Burns in air to form steam.
C. Is a neutral gas, almost insoluble in water.
D. Diffuses more rapidly than carbondioxide.
E. Is prepared by the action of dilute nitric acid on zinc metal.

(vi) Which one of the following substances where its dissolution in water causes permanent hardness of water?
A. Sodium carbonate
B. Magnesium carbonate
C. Calcium sulphate
D. Sodium chloride
E. Calcium carbonate

(vii) What will happen as water is being added to an acid?
A. The acid becomes more acidic and its P^H goes down.
B. The acid becomes neutral and its P^H becomes 7.
C. The acid becomes less acidic and its P^H goes down.
D. The acid becomes more acidic and its P^H goes up
E. The acid becomes less acidic and its P^H goes up.

(viii) A Bunsen burner will produce a yellow flame when;
A. The air hole is fully closed.
B. The gas tap is fully opened.
C. Sufficient gas is supplied to the Bunsen burner.
D. The gas tap is partially opened.
E. The air hole is fully opened.

(ix) Which of the following sets of laboratory apparatus are used to measure volume of water?
A. Thistle funnel, separation funnel and beaker
B. Burrete, pipette and measuring cylinder
C. Crucible, Kipp's apparatus and volumetric flask.
D. Volumetric flask, distillation flask and test tube.
E. Test tube, beaker and gas jar.

(x) Form Four students in a certain school were doing a scientific research about malaria disease in the Coast Region. Which among the following scientific procedures was used to accept or reject the hypothesis made?
   A. Data interpretation
   B. Experimentation
   C. Conclusion
   D. Hypothesis formulation
   E. Problem identification

2. Match the chemical composition of fire extinguisher in **List A** with the corresponding extinguisher in **List B** by writing the letter of the correct response beside the item number in the answer booklet provided.

LIST A	**LIST B**
(i) The extinguisher composed of bromochlorodifluoro-methane.	A. Dry chemical extinguisher
(ii) The extinguisher composed of potassium acetate.	B. Fire blanket extinguisher
(iii) The extinguisher composed of monoammonium phosphate with a nitrogen carrier.	C. APW extinguisher
(iv) The extinguisher composed of fine sodium bicarbonate powder pressurized by nitrogen.	D. Carbon dioxide extinguisher
(v) The extinguisher composed of proteins and fluoro-proteins.	E. Halon extinguisher
(vi) The extinguisher composed of tap water pressurized by air.	F. ABC extinguisher
	G. Sand bucket extinguisher
	H. Wet chemical extinguisher
	I. Foam extinguisher.

SECTION B (54 Marks)

Answer **all** questions in this section

3. (a) Arrange the alkaline earth metals available in the periodic table of the first twenty elements as they appear from the top to the bottom. Explain briefly why the atomic radius increases as one is moving down the alkaline earth metals arranged.

   (b) The Chemistry laboratory is designed for chemical tests. Identify **four (4)** main rules to be observed during these tests for accurate results.

4. (a) A chemical substance L has a density of $1g/cm^3$ at 4^0C, it is neutral to litmus paper, freezes at 0^0C and boils at 100^0C.

   (i) Specify the name and the chemical formula of substance L
   (ii) How can substance L be confirmed in the laboratory? Describe two confirming methods and provide its positive observations.
   (iii) State two main physical properties of substance L and show the usefulness of each property in our daily life.

   (b) With two points, differentiate sodium atom from chlorine atom.

5. (a) Metals are extracted from their ores. Using two important metals in the table below, identify the substance(s) apart from their chief ores which are added during the extraction process and finally, suggest their functions.

Name of the metal to be extracted	**Substance(s) added during extraction**	**Function(s) of the substance(s)**
Sodium		
Iron		

   (b) Consider the reaction between aqueous solutions of silver nitrate and calcium chloride.
   (i) With state symbols, write the balanced molecular equation for this reaction.
   (ii) Give three usefulness of the equation in **(b) (i)** over word equation.

6. (a) Explain each of the following observations.
   (i) Why Methylpropane has a lower boiling point than butane although both have the same molecular mass.
   (ii) Why ignition sources such as smoking are not allowed at petrol stations.

(b) The following results were obtained in an experiment to find the molecular formula of magnesium oxide when magnesium burnt in air.

Mass of crucible = 12.5g
Mass of crucible + magnesium = 14.9g
Mass of crucible + magnesium oxide = 16.5g

(i) What mass of magnesium was used in the experiment?
(ii) How many moles of magnesium atoms were used in the experiment?
(iii) What mass of oxygen combined with magnesium?
(iv) How many moles of oxygen combined with magnesium?
(v) Use the results in (b) (i)-(iv) to write the molecular formula of magnesium oxide

7. (a) The following table contains some names of laboratory apparatus.

Tripod stand	Combustion tube
Measuring cylinder	Retort stand and clamp
Pipette	Wire gauze
Petri dish	Burette
Conical flask	Bunsen burner

Using the names given in the table, identify five apparatus that are used in volumetric analysis and suggest their functions.

(b) What does the statement "a reaction has reached equilibrium" mean with regard to the following?
(i) Forward and reverse reaction rates.
(ii) The amounts or concentrations of the reactants and the products.

8. (a) With the aid of chemical equations, describe how sulphuric acid is produced in large scale using the contact process.

(b) A current of 1.5A was passed through $250cm^3$ of $0.5M\ AgNO_3$ aqueous solution for 1 hour and 10 minutes. The anode was platinum and the cathode was silver.
(i) Give the equations for the reactions which took place at the cathode and anode.
(ii) Describe the changes that would occur if the platinum anode was replaced with a silver anode.

SECTION C (30 Marks)

Answer **two (2)** questions in this section.

9. Describe **six (6)** major changes that matter is likely to undergo in our daily life and state clear examples in each case.

10. When dilute nitric acid is added to a green solid P, a blue solution X is formed and a gas R that forms a white precipitate with limewater. Heating the blue solution of the hydrated salt X, it initially loses its water of crystallization, and then decomposes to form a black solid S, brown fumes of gas Y, and a gas Z that relights a glowing splint.

    (a) Identify the solid X and gases Y and Z
    (b) Write balanced chemical equation for the reaction between the solid P and dilute nitric acid.
    (c) Give a balanced chemical equation for the formation of the solid S.
    (d) Describe two chemical properties of gas Z
    (e) Give the IUPAC name of a green solid P

11. What are the **five (5)** immediate efforts to be done by the National Environment Management Council (NEMC) in Tanzania in order to ensure that land is not polluted?